LOG WEB
Help Manual

Current LOG WEB UI — Revision 2026-08-26

Contents

Introduction

LOG WEB is the browser user interface for daily operation and configuration of the LOG system. This manual describes the current **new** LOG WEB interface and is intended for operators and service users.

Naming Note

LOG WEB is the current product and user-interface name. **Morfeas WEB** is the legacy name. Some internal paths, service names, repository names, protocol identifiers, and code comments may still use `Morfeas`, `Morfeas_WEB`, or related forms for code compatibility and to avoid breaking existing deployments. Operator-facing UI and current user documentation should use **LOG WEB**, except where an exact technical identifier must be shown.

Unless a site-specific deployment says otherwise, open the system in a browser with:

- `http://<device-ip>/`

This manual does not describe the retired legacy Morfeas WEB layout.

Main Page

The landing page is **ISO Channels Linker**. It is the main operational screen.

What You See

The page contains:

- a top bar with the FTP backup directory chip and rolling system summary,
- a search bar,
- action buttons for importing and adding channels,
- the main ISO channel table,
- a right-click context menu for channel actions,
- a menu button that opens all secondary pages.

Main Table

The table shows the configured ISO channels and related runtime information. Typical columns include:

- status,
- ISO channel name,
- device type,
- connection anchor,
- current value,
- history,
- next calibration,
- minimum and maximum values,
- description.

The page refreshes automatically while it is visible, so operators normally do not need to reload manually.

Next Calibration

The **Next Calibration** column is calculated from calibration metadata stored with each ISO channel.

- For **SDAQ** channels, calibration date and period come from the SDAQ runtime calibration data when the device reports it.
- For **IOBOX**, **MTI**, and **NOX** channels, calibration metadata is stored in the OPC UA channel configuration as `CAL_DATE` and `CAL_PERIOD`.
- When a non-SDAQ channel is added and no other calibration metadata is entered, the Add Channel popup defaults `CAL_DATE` to the current date and `CAL_PERIOD` to 12 months.
- The displayed next calibration date is `CAL_DATE + CAL_PERIOD` months.
- A missing or invalid calibration date or period is shown as `--`.

For example, a non-SDAQ channel added on `2026-05-11` with the default 12-month period is shown with next calibration `2027-05-11`.

Measurement Display Rules

The **Value** column follows the runtime measurement semantics of the current system.

- Normal valid measurements are shown as numeric values with the unit when a unit is available.
- **Out of Range** and **Over Range** continue to show the actual runtime measurement value. They are not converted to an error code.
- Reserved runtime error codes are shown as **red integers** in the **Value** column.
- Reserved error codes are shown without units and without decimal formatting.
- Reserved error codes are not collapsed to `--`.

Reserved Measurement Error Codes

The following reserved values may appear in the **Value** column, in OPC UA downstream consumers, and in connected tools that read the same measurement path. These values are emitted by LOG core. The web interface displays them when they are present in runtime data; it does not create these numeric error codes from status text alone.

Code	**Meaning**	**Typical Situation**
`-901`	OFF-Line / Disconnected	Handler is alive, but an individual source, receiver, sensor, or channel is disconnected or intentionally unavailable.
`-902`	No Sensor	The device reports that no sensor is present on that measurement channel.
`-903`	Stall / Heating	SDAQ channel is not receiving fresh samples, or NOX is heating / not ready for valid measurement.
`-904`	Unclassified	Runtime data is invalid, but core cannot map it to a more specific condition.

`-905`	Unreachable / Unregistered	Device-level source is gone or unreachable. Examples: handler/source node missing, IOBOX/MTI Modbus transport failure, or retired MDAQ source.
`-906`	Standby / Heater off	Device is reachable but intentionally not measuring. Currently used for NOX heater-off state.
`-907`	Signal Invalid / Data Invalid	Data is arriving but failed validation, usually a telemetry signal-quality or link-layer problem. Currently used by MTI.

Device Status, Web Value, and OPC UA Value

The same reserved value can have a different operator-facing status depending on the device context. Read **Status** first for the reason, then **Value** for the numeric value sent to downstream consumers.

Web Status below is the linker-table **Status** text. **Web Value** comes from handler logstat files and usually represents interval data. **OPC UA mapped Value** refers to the linked ISO channel **Value** node used by downstream OPC UA clients.

Device	**Web Status / Situation**	**Web Value**	**OPC UA Value**	**Meaning**
SDAQ	Okay	actual value	actual value	Channel is valid and measuring normally.
SDAQ	Out of Range / Over Range	actual value	actual value	Channel reports a real measurement, but it is outside the expected range.
SDAQ	No sensor	`-902`	`-902`	SDAQ reports no sensor on the channel.
SDAQ	Stall	`-903`	`-903`	SDAQ is online, but fresh samples are not arriving.
SDAQ	Unclassified	`-904`	`-904`	SDAQ reports an invalid channel condition without a more specific mapping.
SDAQ	OFF-Line / source missing	`--` when no logstat value is available	`-905` fallback	Linked source node is missing; OPC UA still returns a numeric unavailable value.
IOBOX	Okay	actual value	actual value	Channel is valid and measuring normally.
IOBOX	Unreachable	`-905`	`-905`	IOBOX device-level Modbus transport is unreachable.
IOBOX	Disconnected	`-901`	`-901`	IOBOX handler/device is alive, but the RX/channel path is not responding.
IOBOX	Unconfigured	`--`	`-905` fallback if source node is missing	IOBOX device is reachable, but the linked RX/channel path is not present in runtime logstat. Check IOBOX configuration and wiring.
IOBOX	No sensor	`-902`	`-902`	IOBOX reports no sensor on the channel.

IOBOX	Unclassified	-904	-904	IOBOX reports an invalid condition without a more specific mapping.
MTI	Okay	actual value	actual value	Channel is valid and measuring normally.
MTI	Unreachable	-905	-905	MTI device-level Modbus transport is unreachable.
MTI	Disconnected	-901	-901	MTI handler/device is alive, but telemetry is not responding.
MTI	Radio Disabled / Disabled	-- when logstat has no channel; otherwise -901 if core stamped the retained source	-901 when the retained source node remains; -905 if source node is gone	MTI telemetry was intentionally disabled or reconfigured. Web does not invent a numeric value.
MTI	No sensor	-902	-902	MTI telemetry reports no sensor on the channel.
MTI	Signal Invalid	-907	-907	Packets are arriving, but telemetry data failed validation.
MTI	Error / Unclassified	-904	-904	MTI telemetry is online, but the data is invalid without a more specific mapping.
NOX	Okay	actual value	actual value	UniNOx sensor is active and measuring normally.
NOX	Handler not running, NOX role removed, or sensor never detected	--	-905 fallback if ISO link exists	NOX source node is missing. Web reports missing logstat/runtime source; OPC UA reports missing mapped source.
NOX	OFF-Line after previously detected sensor timeout	-901	-905 if source node was removed	Logstat records the timed-out sensor as offline; OPC UA may only see the missing source node.
NOX	Heater off	-906	-906	Sensor is active, but heater-off state makes measurement unavailable.
NOX	Heating / Not in temperature	-903	-903	Sensor is active, but it is not ready for valid measurement.
NOX	Unclassified / invalid sensor condition	-904	-904	Sensor is active, but the reported measurement is invalid.

MTI Disabled note: Disabled is an operator-selected radio mode, not a measurement fault. If MTI telemetry is disabled, the web shows **Status = Disabled** and leaves **Value** as -- unless LOG core explicitly provides a numeric channel value. This keeps OPC UA/logstat measurement values owned by core and prevents the web interface from inventing error codes.

NOX mapped-channel note: NOX uses dynamic sensor nodes for UniNOx addresses. If the NOX handler is not running, the bus has been switched away from NOX, or a sensor address has never been detected, the web linker table may show **Status = OFF-Line** with **Value =** — because no logstat runtime source exists. The OPC UA mapped **Value** may return -905 for the same ISO link because the mapped source node is missing. These two displays describe the same missing-source condition from different layers.

LOG core treats a detected NOX sensor as active through exactly 10 seconds since its last message. At 11 seconds it is offline. While a sensor is present but reports a heater-off, heating, or invalid condition, core publishes the corresponding reserved value instead of passing through a stale NOx or O2 measurement.

Reading Status Together with Value

Operators should read the **Status** and **Value** columns together:

- **Status** = **Out of Range** or **Over Range** with a numeric **Value** means the device still reports a measurement, but the channel is flagged as outside the expected calibrated or device range.
- **Status** = **No sensor**, **OFF-Line**, **Disconnected**, **Stall**, or **Unclassified** together with a red negative **Value** means the measurement is not valid and the code should be interpreted using the error-code table above.
- **Status** = **Unconfigured** with **Value** = — means the device is reachable, but the configured runtime channel path is not present.
- **Status** = **Disabled** with **Value** = — means the function is intentionally turned off, not failed.

Search

Use **Search Input** to filter channels in the current table.

Add Channel Device Search

The Add Channel popup uses device search to choose runtime source paths for new links. Search results are limited to the selected device family and to paths that are not already linked in the current OPC UA channel configuration.

- **SDAQ**: search lists real channels from detected, registration-complete SDAQ devices that are not already linked. The saved identity is the canonical `serial.CHn`; the current CAN bus/address is display-only and is never saved as channel identity. A real device channel may be linked even when it currently reports **No sensor**.
- **IOBOX**: search lists channels from IOBOX devices whose runtime connection status is **Okay**. A receiver marked **Disconnected** is not listed. Individual channels under an online receiver may still be listed even when their current status is **No sensor** or otherwise invalid, matching the current channel-selection contract. Each IOBOX receiver also exposes `RXn.Status` and `RXn.Success` link metrics. `RXn.Status` reports `1` when the receiver link is okay and `0` when it is not okay, with no unit. `RXn.Success` reports the receiver packet success ratio from `0` to `100%`. If the IOBOX device itself is unreachable after an ISO channel has already been linked, these mapped values follow the normal missing-source error path and may report `-905`.
- **MTI**: search lists channels only when the MTI runtime connection status is **Okay** and the radio/telemetry mode provides channel data. Modes **TC4**, **TC8**, **TC16**, and **QUAD** expose their channel list. **RMSW/MUX** exposes detected **Mini_RMSW** measurement channels. **Disabled** does not expose Add Channel candidates.
- **NOX**: search lists **NOx** and **O2** entries for detected UniNOx sensor objects. The current measurement does not need to be valid; for example, heater-off, warming-up, or previously detected **OFF-Line** sensors may still be linked and controlled later. OFF-Line historical candidates are marked with `(OFF-Line)` in the search result.

The search popup and save validation both enforce device-family matching. IOBOX candidates are not accepted as MTI links, and MTI candidates are not accepted as IOBOX links, even if the physical equipment connected behind the device can be similar.

Typing in the Path field only searches the current candidate list. A typed value must resolve to an available candidate before Save; it is not a Manual or offline channel-entry path. SDAQ Unit is read-only and comes from live device metadata, so Add does not store an SDAQ Unit in the OPC UA XML configuration.

Add Channel ISO and Multilinking Rules

With **Range = 1**, the entered ISO code may be saved even when it is not present in the selected ISOstandard.

With **Range greater than 1**, every generated base ISO code must exist in the selected ISOstandard and remain in the same measurement category. The numeric suffix is expanded without adding an extra leading zero; for example, `TEMP099` continues as `TEMP100`.

A selected cylinder postfix is applied only after the batch ISO sequence is validated. For example, the ISOstandard lookup uses `TEMP1`, `TEMP2`, and `TEMP3`, while cylinder 5 is saved as `TEMP1_5`, `TEMP2_5`, and `TEMP3_5`.

The multilinking naming and rejection rules follow *Rules for how to name channels in Laboratory measurement systems*, Document ID `DMTA00018532`.

Range Add is one all-or-nothing operation. LOG WEB rechecks every selected channel immediately before writing; if any source, ISO name, or identity conflicts, no channel in the range is written.

SDAQ Link and Gateway Mapping

Adding a channel in LOG WEB and mapping it in the Lab OPC UA Gateway Configuration Tool are separate operations. LOG WEB creates the link; the Gateway discovers it during browse and stores its own metadata snapshot when the point is configured and saved.

Online Add and First Mapping

For a new point:

1. In LOG WEB, confirm the SDAQ channel is online and that Unit, device Type, CAN location, and calibration information are correct.
2. Add the channel from the current Unlinked search results.
3. In the Gateway Configuration Tool, perform a complete browse or rebrowse of the LOG endpoint.
4. Confirm the Lab OPC UA Gateway mapper shows the expected Unit, Type, Address, and SDAQnet values.
5. Configure and save the point.

Device Goes Offline Before First Mapping

The XML link and ISO object remain, but Current Core hides the required Unit browse reference while current runtime metadata is incomplete. A fresh Gateway browse therefore should not offer the point for first mapping.

Do not force **Configured** or save empty or uncertain metadata. Reconnect the device, wait for registration and metadata to complete, confirm the values in LOG WEB, and then perform a complete rebrowse before the first Configure/Save.

Restored Historical Channel Is Currently Offline

Local JSON or FTP Restore may restore a valid canonical `serial.CHn` identity while that SDAQ is offline. This preserves the link so the same serial and channel can recover automatically if it later appears on this LOG; it does not prove that the device is currently detected, and it does not configure the Gateway.

Before the device is ready, a fresh Gateway browse should not map the point. After the same serial/channel is online and its metadata is correct, perform a complete rebrowse and, if the point has never been configured on this endpoint, configure and save it. A serial that never appears on this LOG remains offline and is never silently moved to another address or device.

Previously Mapped Device Goes Offline

Do not delete an existing saved Gateway point merely because a fresh browse temporarily shows it as missing or not mappable. The Gateway keeps its saved metadata and Value NodeId. Value may become `-905` and status may show offline/unregistered; when the same `serial.CHn` returns to the same LOG, Value recovers without remapping.

If the SDAQ is moved to another LOG, the old LOG mapping remains offline. Add the device from the new LOG's Unlinked list and perform first mapping against the new LOG endpoint.

Browse Cache, No Sensor, and Unclassified Unit

The Core browse gate affects new browse results; it cannot clear a point tree already cached by the Gateway Configuration Tool. A device can go offline after browse but before Save, so confirm that it is still online and perform the tool's complete browse/rebrowse operation immediately before first Configure/Save. A normal page refresh is not guaranteed to clear the Gateway's internal browse cache.

The Gateway does not automatically discover a new ISO object and does not automatically refresh metadata it has already saved. After a sensor or Unit changes, rebrowse, verify the metadata, and save again.

No sensor means the physical SDAQ channel exists but has no sensor connected; it may be linked and typically reports `-902`. **Unclassified** Unit means Core received a unit code it cannot classify; it is not an empty string, but it is not confirmed engineering metadata. Link success does not mean a point is ready for first Gateway mapping. Connect the actual sensor and confirm Unit, Type, Address, SDAQnet, and calibration before first mapping. If `Unclassified` was already saved in Gateway metadata, later sensor recovery does not rewrite it automatically; rebrowse and save again.

Keyboard Shortcuts

The following keyboard shortcuts and key actions are currently implemented in LOG WEB.

Main ISO Channels Linker

- **Delete**: delete the currently selected channel rows. A confirmation dialog is shown first.
- **Ctrl+Z**: undo the most recent successful delete. Only one delete step is remembered.

- **Escape**: close the main menu or an open column filter menu.

Add Devices Popup

- **Ctrl+S**: save the device currently being added when the add form is open.
- **Delete**: remove the currently selected removable devices.
- **Escape**: close the popup window.

Add Channel Popup

- **Ctrl+S**: save the current Add Channel form.
- **Enter** in the ISO field: accept the typed ISO code as a custom manual ISO value.
- **Enter** in the sensor path field: select the typed device path only when it matches a currently available search-pool candidate. It cannot create a channel outside that list.
- **Escape**: close the popup window.

Edit Link / Replace Link Popup

- **Ctrl+S**: save the current Edit or Replace form.
- **Escape**: close the popup window.

Other Popup Pages

- **Escape**: close popups such as Advanced Settings, Backup & Restore Channels, Check System Status, Network Configuration, Version, and system control pages.

Menu Overview

The main menu is grouped as follows.

Menu Item	Purpose
Network Configuration	Edit hostname and Ethernet IPv4 settings.
Add Devices	View current devices and add or remove supported devices.
Backup & Restore Channels	Configure FTP backup target, create backups, and restore saved bundles.
Check System Status	View live runtime details, log files, and system journal output.
Advanced Settings	View MAC/CAN information, configure NTP server, and manage ISOstandard XML.
System Update	Check for available system updates, then update LOG WEB and LOG core when desired.
System Reboot	Reboot the device.
System Shutdown	Power off the device.
Version	Show current web and core version information.
Help	Download this manual as PDF.

Network Configuration

Open **Network Configuration** from the menu to edit hostname and Ethernet settings.

Fields

The page contains:

- **Hostname**
- **Mode**: Static or DHCP
- **Network IP**
- **Gateway**
- **DNS addr**

Preset buttons are available at the top of the page for quick loading of predefined values.

Buttons

- **Reset**: restore the form to the configuration loaded when the popup was opened or last refreshed.
- **Set**: apply the current values.
- **Close**: close the popup window.

Operator Notes

- If you change the device IP address, reconnect the browser to the new HTTP address after the change is applied.
- DHCP mode disables manual IPv4 entry.
- Presets are convenience values only. Review them before applying.
- **Reset** only restores the values already loaded into the current popup. It does not reload from the device, roll back a saved IP change, or reconnect the browser.
- After an IP change, or after the old connection is interrupted during a static IP switch, the current page may be stale. Open the new device address before making more network changes.

Example Message (Progress):

> Applying configuration: validating and writing system network settings...

Example Message (Apply Success):

> Apply completed successfully. mode=<mode>, effective ip=<device-ip>.

Example Message (IP Changed):

> Apply completed. IP changed from <current-device-ip> to <new-device-ip>.
> Current page may disconnect by design. Open http://<new-device-ip>/.

Example Message (Switch Interrupted):

> IP switch result is not confirmed yet because the old connection was interrupted.
> Try http://<new-device-ip>/ first. If still unreachable, wait 20–30 seconds and try http://<new-device-ip>/ again.

Example Message (Failure):

Apply failed: <error details>

Add Devices

Open **Add Devices** to manage the current device inventory.

Page Layout

The page has three main areas.

- **CAN Bus Setup**: shows each CAN bus, current mode, bitrate, state, detected device count, and available actions.
- **Configured Devices**: shows XML-backed devices that can be removed or configured.
- **Detected Devices**: shows auto-detected runtime devices, especially SDAQ devices found from logstat data.

The page refreshes while the popup is focused. Polling pauses during remove or role-change operations so the displayed state does not fight the operation in progress.

SDAQ Operation Flow

- SDAQ devices are auto-discovered from runtime `logstat` data.
- They appear in **Detected Devices**; they are not manually added with the **Add...** form.
- To use a CAN bus for SDAQ, use the **Switch to SDAQ** action in **CAN Bus Setup**. The system sets the SDAQ role and expected bitrate, restarts the LOG service, then waits for SDAQ runtime data to appear.
- After SDAQ devices are detected, link individual SDAQ channels from the main page with **Add Channels**.
- If a bus is in SDAQ mode but no SDAQ appears, check wiring, power, and CAN bitrate/state in the same table.

LOG core remembers the address associated with an SDAQ serial number for 14 days after the device goes offline. A returning device normally recovers that address. ISO channel identity is still the canonical `serial.CHn`; the remembered CAN address is only an advisory runtime reservation and is not the saved channel identity. After an on-card software upgrade from the older address-cache format, the cache is rebuilt from devices currently online, so addresses may be reassigned without changing canonical channel identity.

NOX Operation Flow

- NOX is managed by CAN bus role, not by the manual **Add...** form.
- Use **Switch to NOX** on the target CAN bus to assign that bus to NOX operation. The transition changes the role/bitrate together and restarts the LOG service.
- When a bus is in NOX mode, **Open NOX Config** opens the NOX configuration page for that bus.
- Use the NOX configuration page for sensor setup and NOX-specific live controls.

- To remove NOX from a bus, use **Switch to SDAQ**. When removing a configured NOX row, the UI may also ask whether to restore the bus to SDAQ.

IOBOX Operation Flow

1. Click **Add...**.
2. Select **IO-BOX**.
3. Enter **Device Name**, for example `IOBOX_01`.
4. Enter the IOBOX IPv4 address.
5. Click **Save**.
6. Confirm the service restart warning.
7. Wait for the page to refresh and show the configured device runtime status.

Behind the UI, saving an IOBOX writes the device into the LOG configuration XML and restarts the LOG service. Duplicate device names and duplicate IPv4 addresses are blocked.

MTI Operation Flow

1. Click **Add...**.
2. Select **MTI**.
3. Enter a **Device Name**; for MTI it must follow D-Bus element rules: letters, numbers, and underscore only, and it must not start with a digit.
4. Enter the MTI IPv4 address.
5. Click **Save**.
6. Confirm the service restart warning.
7. After the device is listed, use **Open MTI Config** from the configured row to manage MTI setup, telemetry mode, and MTI-specific controls.

Behind the UI, saving an MTI writes the device into the LOG configuration XML and restarts the LOG service. The page compares configured name/IP with runtime MTI logstat data to show whether the device is connected.

Remove Devices

- Select one or more rows in **Configured Devices**.
- Click **Remove** or press **Delete** while focus is not inside a form field.
- Confirm the service restart warning.
- The selected XML-backed devices are removed and the LOG service is restarted.

Operator Notes

- SDAQ rows in **Detected Devices** are auto-discovered and are not removable from this page.
- Manual add currently supports IOBOX and MTI only. SDAQ is auto-discovered; NOX is controlled through CAN bus role switching.

- If the page shows a legacy MDAQ warning, add/remove and CAN role actions are blocked until the legacy MDAQ XML configuration is removed manually.
- Any add, remove, or CAN role transition may need 10–60 seconds because the LOG service restarts.

Backup and Restore Channels

Open **Backup & Restore Channels** for FTP-based backup and restore.

FTP Server Setup

The page contains:

- **Host IP**: FTP server IPv4 address. The default expected server is `10.193.135.70`.
- **Engine Number**: remote directory name for this engine/device group. It may contain letters, numbers, dot, underscore, and hyphen.
- **Connect & Fetch**: save the FTP configuration, test the FTP login, and load available backup files.
- **Disconnect**: remove the saved FTP configuration and disable automatic backups.

FTP credentials are read from the device-side credential file and are not typed in the UI. When the connection is saved, the main page FTP chip shows the active backup directory.

What Is Backed Up

Each backup bundle contains the current OPC UA channel configuration XML and the LOG device configuration XML. The bundle is compressed, uploaded to the selected FTP engine directory, and stored as an `.mbl` file.

Automatic Nightly Backup

- The deployment installs a root cron job that runs the FTP backup script every night at `00:10`.
- Automatic backup only runs when a valid FTP configuration exists.
- If no FTP configuration exists, the backup script logs that no valid engine number is configured and does not change system state.
- After a backup, the script uploads the backup logs to the same FTP engine directory when possible.

Manual Backup

1. Open **Backup & Restore Channels**.
2. Confirm or enter the FTP server IP.
3. Enter the engine number.
4. Click **Connect & Fetch**.
5. Wait for **Connected to FTP... Configuration saved**.
6. Click **Backup Now**.

7. Wait for the success message and confirm that the new `.mbl` file appears in the list.

Restore

1. Connect to the FTP server and engine directory.
2. Select one backup file from the list.
3. Click **Restore Selected**.
4. Review the preflight errors and warnings. Hard errors must be corrected; warnings require explicit confirmation.
5. Confirm Restore and wait for the result message.
6. Reload the current LOG WEB and verify both channel and device configuration.

- Restore downloads the selected `.mbl` file from FTP.
- Preflight validates both configuration files against the DTD, strict channel/device rules, and IOBOX/MTI channel-to-handler relationships. A hard error prevents all writes.
- Restore is a complete replacement of both the OPC UA channel configuration XML and the LOG device configuration XML; it is not a channel merge.
- The two files are locked and replaced in a fixed order. This prevents ordinary Web operations from interleaving, but it is not a power-loss-atomic transaction across both files.
- If the browser, Web process, or power is interrupted during Restore, run the same Restore again and recheck devices and channels.
- A restored offline SDAQ follows the Gateway mapping rules in **SDAQ Link and Gateway Mapping**. Do not force first Gateway configuration while the device identity or runtime metadata is unavailable.
- Backup files are listed newest-first, and old remote backups are pruned so the engine directory keeps up to 50 `.mbl` files.

Operator Notes

- Backup and restore both require a reachable FTP server.
- If the server is unavailable, the page shows an error and no configuration is changed.
- If two browser sessions edit the FTP configuration, the page detects the updated configuration and refreshes the backup list.

Example Message (Connect Success):

> Connected to FTP at 10.193.135.70. Configuration saved.

Example Message (Connect Failure):

> Connection failed: FTP connect failed

Example Message (Backup Success):

> Backup complete

Example Message (Restore Failure):

> Restore failed: <error details>

Check System Status

Open **Check System Status** to inspect runtime data and logs.

Tabs

The page has three tabs.

Details

This tab shows live system and bus information in tables. Depending on the device and services, values may include:

- CPU temperature,
- uptime,
- bus utilization,
- bus error rate,
- detected SDAQ count,
- incomplete SDAQ count,
- bus voltage,
- bus amperage,
- shunt temperature,
- last calibration timestamps.

Temperature values shown by the current web interface are displayed in degrees Celsius. This includes CPU temperature, SDAQnet shunt temperature, IOBOX/MTI temperature channels, and NOX page shunt temperature.

System Logs

This tab is used for LOG application logger files.

- **Preview Logger**: choose one log file to preview.
- **Reload**: reload the logger list.
- **Export Logs**: select one or more log files and export them as one combined text file.

The preview area shows the selected file content in a terminal-style viewer.

System Journal

This tab is used for journal output from system services.

- **Scope**: choose all services, LOG system service, Apache2, or SSH. The current UI label for the LOG system service may still appear as **Morfeas System** until that compatibility label is renamed.

- **Lines**: choose how many lines to load.
- **Reload**: reload the journal view.
- **Export Journal**: export the currently requested journal text.

How to Use the Tabs

Details Tab

Use this tab first when checking current runtime health. If a service is active but values look stale, compare with the Logs or Journal tabs.

System Logs Tab

Use this tab for LOG component logs when you want application-level detail such as handler messages, update logs, backup logs, or runtime component reports. To export logs, choose one or more files in the **Export Logs** multi-select, then click **Export Selected**. LOG WEB downloads the selected logger files as one combined text file. Use **Reload** if the file list looks stale.

System Journal Tab

Use this tab when the problem looks system-level rather than application-level, for example:

- service startup failures,
- Apache problems,
- SSH availability,
- reboot or shutdown investigation.

To export journal output, choose the **Scope**, choose the number of **Lines**, click **Reload** if needed, then click **Export Journal**. The browser downloads the currently requested journal text as a `.txt` file.

Advanced Settings

Open **Advanced Settings** for machine information, NTP, and ISOstandard management.

Machine Identifier

Displays the MAC address.

SDAQNet (CAN-IFs)

Displays the current CAN interface information for `can0` and `can1`. These values are informational and read-only on this page.

Time Synchronization

The page shows the current NTP server and allows entering a new IPv4 NTP server. Use **Apply** to store the new value.

Example Message (Apply Success):

Applied NTP server: 192.168.1.10

Example Message (Validation Error):

NTP server must be a valid IPv4 address.

ISOstandard

This section lets the user:

- view the currently selected ISOstandard source,
- select a different available ISOstandard file,
- upload a new ISOstandard XML file,
- review the loaded ISO point definitions in a table.

System Update

Open **System Update** to manage full system software updates.

Current Behavior

The page supports:

- checking whether an update is available,
- applying the update,
- deferring the update,
- reloading the page after service recovery.

Reminder Dot

A red dot on the menu item indicates that the system has recorded an available update. The reminder stays visible until a later check confirms that no update is pending, or until a successful update clears it.

Important Note

The current update flow handles the full system update path shown in the progress messages. It checks and updates LOG WEB, applies web deployment steps, checks and downloads LOG core changes, synchronizes the recorded `sdaq-worker` source, builds and installs both `SDAQ_worker` and LOG core when needed, runs core health checks, restarts services, and waits for the web service to become reachable again.

Example Message (Update Available):

Update available. Click "Update Now" to apply, or "Update Later" to postpone.

Example Message (Update Success):

Update flow completed and web service is reachable. Refresh page (Ctrl+F5) to load latest files.

Example Message (Update Failed):

> System update failed: <error details>

System Reboot and System Shutdown

The system control pages provide single-confirmation actions for reboot and shutdown.

System Reboot

Use **Reboot Now** when you want to restart the device and return to service after boot. The page shows post-action guidance so the operator knows when to reconnect and refresh the browser.

Example Message (Reboot Accepted):

> Reboot command accepted. Wait about 60–90 seconds, then open the device URL again and refresh the browser.

System Shutdown

Use **Shutdown Now** when you want to power off the device. After shutdown, the device must be powered on again manually before the web interface becomes available.

Example Message (Shutdown Accepted):

> Shutdown command accepted. Device will go offline. Power it on manually when needed.

Version

The **Version** page shows version and commit information for:

- the LOG WEB repository,
- the LOG core repository,
- the related commit dates.

Use this page when confirming what software is installed on a device.

Import Channels

Use **Import Channels** on the main page to create channels from a JSON file.

How Import Opens

Clicking **Import Channels** opens the browser file picker. Select a `.json` file; there is no separate import popup page.

Expected File Format

The file must contain a JSON array. Each entry must include:

- `ISO_CHANNEL`

- `INTERFACE_TYPE`
- `ANCHOR`
- `DESCRIPTION`
- `MIN`
- `MAX`

Optional fields include alarm fields, `UNIT`, `CAL_DATE`, and `CAL_PERIOD`. For SDAQ imports, LOG WEB ignores `UNIT`, `CAL_DATE`, and `CAL_PERIOD` because these values come from the live SDAQ runtime/calibration path.

Typical Import Flow

1. Prepare or export a JSON array of channel entries.
2. Click **Import Channels**.
3. Select the JSON file.
4. LOG WEB preflights every row and reports **Ready to restore**, **No change**, **Update metadata**, **Invalid**, or **Conflict**.
5. Correct every Invalid or Conflict row, then run preflight again. Core-valid orphan or device-type relationship findings are shown separately as warnings.
6. Review every warning and confirm the operation. Confirmation explicitly acknowledges the current warning list. LOG WEB re-parses and rechecks the complete candidate under the configuration lock and writes the resulting channel set once.
7. The main table refreshes after the atomic merge succeeds.

Validation and Conflicts

- The import fails if the file is not valid JSON or is not a JSON array.
- Local JSON Restore is an atomic channel-set merge: rows in the file are added or updated, but existing channels absent from the file are not deleted.
- Same ISO plus the same source may update allowed metadata. Same ISO with a different source, or the same source with a different ISO, is a conflict.
- Duplicated ISO names or source identities within the file are conflicts.
- If any row is Invalid or Conflict, the whole import writes nothing. There is no partial-success state and no need to guess which rows were changed.
- A warning does not make the file invalid. For example, a valid IOBOX or MTI channel may refer to a handler absent from the current device configuration. Such a channel can be restored but remains offline until the matching handler is added. Warnings must be explicitly acknowledged; a commit without acknowledgement writes nothing.
- Commit checks that the configuration has not changed since preflight, then repeats parsing and conflict checks while holding the lock.
- A canonical offline SDAQ row may be restored, but it is not ready for first Gateway mapping until that serial/channel is online with complete metadata.

Export as Import Source

The right-click **Export** action creates a JSON file in the same structure expected by import. This is the safest way to move selected channel definitions between devices, then review the file before importing it elsewhere.

Right-Click Channel Options

Right-click one or more channel rows in the main table to access channel operations.

Selection Rules

- With one selected row, LOG WEB can show row-specific actions.
- With two or more selected rows, LOG WEB shows only **Delete** and **Export**.
- If no row is selected and you right-click a row, that row is selected automatically.
- If several rows are already selected and you right-click one of them, the multi-row selection is kept.

Action Availability by Scenario

Scenario	**Available actions**
Single SDAQ row	Edit, Delete, Calibration, Export.
Single SDAQ-I or SDAQ-U row	Edit, Delete, Scale, Calibration, Export.
Offline SDAQ row	Replace is added.
Offline complete SDAQ-TC16 group	Replace TC16 is added when the selected source belongs to a full CH1–CH16 offline TC16 group.
Single IOBOX row	Edit, Delete, Export.
Single MTI row	Edit, Delete, Export.
Single NOX row	Edit, Delete, Export.
Multiple rows	Delete, Export.

Edit

1. Right-click one channel row.
2. Choose **Edit**.
3. Review the locked source type, ISO code/postfix, and device path.
4. Change description, min/max, alarm low/high and their values. Unit and applicable XML calibration fields are editable only for non-SDAQ channels.
5. Click **Save**.

Edit keeps type, anchor/path, ISO code/postfix, and build identity unchanged. SDAQ Unit and SDAQ runtime calibration fields are read-only; use the Calibration or Scale page to write SDAQ device calibration.

Delete

1. Select one or more rows.
2. Right-click and choose **Delete**, or press **Delete**.

3. Confirm the delete prompt.

After a successful delete, one undo step is remembered for a short time. Press **Ctrl+Z** to restore the last deleted channel set when there is no conflict with existing live channels.

Export

1. Select one or more rows.
2. Right-click and choose **Export**.
3. The browser downloads a timestamped `LOG_ISOChannel_Linker_Export_Selection` JSON file.

The exported JSON can be used later with **Import Channels**. SDAQ export does not include XML-owned `UNIT`, `CAL_DATE`, or `CAL_PERIOD`; those values are recovered from SDAQ runtime/calibration data when the device is available.

Scale

Scale is available only for **SDAQ-I** and **SDAQ-U** rows.

1. Right-click an SDAQ-I or SDAQ-U row.
2. Choose **Scale**.
3. Review **Current Measurement Value**.
4. Enter **Measurement Input Low**, **Measurement Input High**, **Engineering Output Low**, **Engineering Output High**, and **Engineering Unit**.
5. Review **Scaled Output Value**.
6. Click **Save**.
7. Use **Continue in Calibration** when you want to add or review calibration points for the same channel after saving the quick scale.

The scale popup writes a new two-point calibration table into the selected SDAQ channel calibration data. Scale is itself a calibration operation, not a display-only scaling shortcut. If the channel already has an active point table, LOG WEB shows the existing point count and requires confirmation before replacing it. The new table receives today's calibration date and a re-calibration period of zero; it never inherits the previous table's date, period, or coefficients.

Scale is nominal range setting, not a traceable recalibration. It applies the sensor's declared transfer function (for example, 4–20 mA to an engineering range) and does not use a reference standard. Therefore its fixed period of zero means the channel does **not** appear in the Next Calibration schedule. If the channel requires a traceable calibration interval, use the full Calibration workflow with a reference standard instead of Scale.

Scale may be performed again later. Each confirmed Save is a new calibration and replaces the previously active table with the newly entered two-point table. Save the Scale before choosing **Continue in Calibration**; that button opens the full editor for a subsequent calibration and does not transfer unsaved Scale fields.

Calibration

Calibration is available for SDAQ rows.

Purpose

The calibration popup is a channel calibration point editor for the selected SDAQ channel. It is used to read, inspect, preview, and save the device calibration table.

Page Layout

The popup contains:

- **Upper Section**: device information, calibration metadata, and summary panel share the top area.
- **Current Preview**: `Measured raw value`, `Current Device Output`, and `Preview Output`. Each value has a copy button when the value is available.
- **Point Table**: `Uncalibrated value`, `Reference value`, `Offset`, `Gain`, `C2`, and `C3`.
- **Preview Columns**: `Calibrated value` and `Difference (Calibrated - Reference)`.
- **Summary Panel**: calibration type, slope or gain summary, offset, reference minimum and maximum, and unit.

Buttons and Validation

- **Read from SDAQ**: reload calibration directly from the device.
- **Discard Changes**: restore the last data loaded into the popup.
- **Edit Calibration Points**: unlocks point editing for a channel that already has calibration data. Confirming this action converts the editable table to auto-linear points and may overwrite existing coefficients.
- **Save to SDAQ**: validate the current auto-linear point form and write it to the device.
- Independent date/period-only editing is not available. Calibration Date is stamped automatically with the current date when the point table changes; Re-Calibration Period is saved only as part of that same point-table calibration.
- **CH's Unit** must be selected from the supported runtime SDAQ unit list.
- **Used Points** must be an integer between zero and the device maximum.
- In point editing mode, `Uncalibrated value` and `Reference value` must contain finite numbers.
- `Uncalibrated value` (device `Measure`) must be strictly increasing by row after conversion to the device's float32 representation. LOG WEB does not reorder rows. Values that look different as decimal text but become the same float32 value are rejected.
- `Reference value` may increase or decrease, depending on the sensor and engineering range.
- A new zero-to-one-point calibration uses unity gain and calculates an offset. Re-saving an existing genuine linear one-point table may retain that table's gain. Reducing a multi-point or polynomial table directly to one point is rejected because no valid slope can be inferred.
- A one-point table has no bounded calibrated interval, so the SDAQ Out-of-Calibrated-Range supervision is unavailable. LOG WEB requires a separate warning confirmation before saving one point.

`Offset`, `Gain`, `C2`, and `C3` are shown for transparency. In normal point editing mode they are read-only and LOG WEB recalculates them from the calibration points before saving. The goal of each used point is that `Difference (Calibrated - Reference)` is as close to zero as possible.

After a Calibration Write

After a successful Scale or Calibration save, use **Read from SDAQ** once and verify the active point count, point values, unit, calibration date/period, and live output before starting the formal measurement run. Normal operation does not require repeated writes, a long polling window, or a mandatory power cycle.

If the table or output is abnormal, stop using that channel, preserve the readback evidence, and power-cycle the target SDAQ itself before reassessing it. Restarting only the browser, LOG service, or `SDAQ_worker` is not the same as removing power from the SDAQ. The recorded delayed-change event has no confirmed root cause, so LOG WEB does not add repeated writes or artificial stress operations to reproduce it.

Replace

Replace is shown only for offline SDAQ rows. It uses the Edit Link popup in replace mode.

1. Right-click the offline SDAQ channel you want to replace.
2. Choose **Replace**.
3. The popup opens with the source type locked.
4. Use device search to choose the replacement source path.
5. Review the locked ISO code, description, min/max, alarms, and replacement identity.
6. Click **Save**.

Replace changes the source identity for an existing offline SDAQ link while preserving its channel description, limits, and alarm settings. The replacement must be selected from the current Unlinked device list; an address cannot be typed or invented. SDAQ `UNIT`, `CAL_DATE`, and `CAL_PERIOD` are removed from channel XML because these values are owned by the replacement SDAQ runtime/calibration data. Family mismatch is blocked, and both source and target SDAQ subtype must be known and equal.

Example Message (Blocked Replace):

> Source SDAQ subtype is unknown; choose a channel whose type can be verified.

Example Message (Blocked Replace):

> Subtype mismatch: expected SDAQ-TC1, got SDAQ-AI8.

Replace TC16

Replace TC16 is shown only when the selected source belongs to a resolvable offline SDAQ-TC16 group that is complete as channels CH1 through CH16.

1. Right-click any channel in the full offline TC16 source group.
2. Choose **Replace TC16**.
3. LOG WEB selects all 16 source channels.

4. The Replace TC16 popup shows the source group summary and channel mapping.
5. Choose one eligible unlinked TC16 target device.
6. Confirm the operation.
7. All 16 source channels are remapped to the target device in one atomic operation.

- Replace TC16 is all-or-nothing.
- If any validation fails, no channel mapping is changed.
- Target selection is at device-group level, not one channel at a time.
- If no eligible target is listed, there is currently no fully free TC16 target device available.

Example Message (Success):

> Replace TC16 complete: 16 channels updated.

Example Message (Failure):

> Failed to apply TC16 replace. [tc16_target_not_unlinked]

Troubleshooting

This section gives quick operator guidance for common problems.

The Web Page Does Not Open

- Confirm that you are using the current device IP address over HTTP.
- If the IP was recently changed, reconnect to the new address.
- If the device was rebooted or updated, wait for services to come back and refresh the browser.
- If still unavailable, check **System Journal** or use SSH to verify Apache and LOG services.

FTP Backup Cannot Connect

- Check that the FTP server IP is reachable from the device.
- Confirm that the engine number is correct.
- If the page says configuration was removed, reconnect first.
- If the internal FTP server is currently unavailable, backup and restore cannot be tested end-to-end.

No Backup Files Are Listed

- Verify that the selected engine directory contains backup files.
- Confirm that the FTP configuration belongs to the expected engine number.
- Reload after reconnecting to the FTP server.

Replace or Replace TC16 Option Is Missing

- Confirm that the selected row supports that replace mode.
- For Replace TC16, confirm that the source is a complete offline TC16 group with channels CH1 through CH16.
- If the source group is partial, use normal single-channel replace instead.

No Eligible TC16 Target Is Listed

- No full unlinked TC16 target device is currently available.
- If channels on the target are already linked, the target is blocked.
- Refresh the popup after any device or link changes.

Logs or Journal Look Empty

- Use **Reload** once to refresh the selected tab.
- Verify that the correct logger file or journal scope is selected.
- If the service has not produced messages yet, the view may be temporarily empty.

Red Negative Measurement Codes Are Shown

- `-901` means the handler is alive, but the individual source, receiver, sensor, or channel is offline or disconnected.
- `-902` means no sensor is present.
- `-903` means the channel is stalled, heating, or not ready for valid measurement.
- `-904` means the runtime data is invalid without a more specific mapping.
- `-905` means the device-level source is unreachable or the mapped OPC UA source node is unregistered.
- `-906` means standby / heater off.
- `-907` means signal invalid / data invalid.
- When these codes are shown, the measurement should not be interpreted as a physical live value.

Restored SDAQ Point Cannot Be Mapped Yet

Confirm that the stated serial is connected to this LOG, registration is complete, and the channel is online. Check Unit, Type, Address, SDAQnet, and calibration. When these values are ready, perform a complete Gateway rebrowse before first Configure/Save. A restored XML link by itself is not proof that the device is present.

Previously Mapped Point Appears Missing After SDAQ Went Offline

Do not immediately delete the saved Gateway point. First determine whether the SDAQ is temporarily offline or has moved to another LOG. A same-serial return to the same LOG restores the existing Value path; moving it to a new LOG requires Add and first mapping on the new endpoint.

Gateway Shows the Old Unit After a Sensor Change

Gateway metadata is a saved snapshot and is not refreshed automatically when SDAQ runtime metadata changes. Reconnect the sensor, confirm its current Unit in LOG WEB, then perform a complete Gateway rebrowse and configure/save the point again.

Restore Was Interrupted

Run the same FTP Restore again, then reload LOG WEB and verify both Devices and ISO Channels. Treat restored offline SDAQ channels according to the first-mapping guidance above.

FTP Restore replaces the saved XML files as a bundle and does not rewrite individual SDAQ channel fields through the Local JSON rules. A historical FTP backup may therefore still contain SDAQ `UNIT`, `CAL_DATE`, or `CAL_PERIOD`; current Core ignores those XML fields for SDAQ and uses live device data. Local JSON Restore instead reports those fields as ignored and removes them from the resulting SDAQ channel XML.

Import Reports an Invalid or Conflict Row

The entire Local JSON import was rejected and no rows were written. Correct or remove every reported row, run preflight again, and then confirm the whole merge. There is no partial result to clean up.

If preflight reports only warnings, read them before continuing. The file remains eligible to commit, but LOG WEB requires explicit confirmation. Cancelling or submitting without warning acknowledgement writes nothing.

Scale Reports an Existing Calibration

This is informational, not a permanent block. Scale is a new two-point calibration. Review the current active point count and the replacement warning, then confirm only if the old point table should be replaced. The new Scale receives today's date and period zero. It is a nominal range setting and will not appear in the Next Calibration schedule. Choosing **Continue in Calibration** opens the full editor for another calibration; it does not save unsaved Scale entries.

SDAQ Calibration Readback or Output Is Abnormal

Stop using the affected channel for measurement and preserve the table/readback evidence; do not attempt rapid repeated writes. Power-cycle only the affected SDAQ, wait for the same serial/channel to return, then read and check the table and output again. Escalate the device for investigation if the values do not recover or the condition returns.

Channel Configuration Validation Is Unavailable

LOG WEB deliberately refuses Add, Edit, Replace, Delete, Import, or Restore writes when the delivered `Morfeas.dtd` cannot be read. This prevents an unvalidated `OPC_UA_Config.xml` from replacing the working configuration. The DTD must be installed in `/home/morfeas/configuration/`, beside `OPC_UA_Config.xml`. Restore the matching product DTD and its read permissions, then retry the original operation. Do not create a blank or locally modified replacement. Existing Core channels are not deleted merely because this Web validation dependency is unavailable.

Update Reminder Dot Is Still Visible

- The reminder is based on the stored update flag.

- Run **Check Again** in the System Update page to refresh the latest status.
- After a successful update or a clean re-check, the reminder should clear.

Help PDF Maintenance

The Help menu downloads a PDF file from:

- `docs/help_manual.pdf`

The maintained source for this file is located under:

- `Morfeas_WEB/docs/manual/`

This keeps the active user manual next to the current LOG WEB code instead of inside the archived legacy documentation tree.

Legacy Documentation Status

The old documentation tree under `LOG-web/Docs/Morfeas_WEB_Docs` is retained only as historical reference. Its structure and API descriptions belong to the legacy Morfeas WEB implementation and should not be used as the active operator manual for the current LOG WEB UI.